Kerbo-S Draft Test Plan, Sprint 1

AI-Assisted Kerberos Security Auditor | CSCE 4907 Senior Capstone

GitHub URL: https://github.com/chancellorr/Cybersecurity-Capstone.git

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Purpose

This plan defines 57 test cases that cover every Must requirement in Requirements Document v2.0, each with steps, an expected result and a status. No tests have been run yet, so every case is Not Yet Tested. No template was supplied for testing, so the structure follows the Sprint 1 submission requirements: what is tested, cases with expected results and steps, status, and known issues.

1. What Will Be Tested

In scope: data collection and parsing, the eleven audit rules (KC-01 to KC-11), risk scoring, database storage, AI Gateway safety and fallback behavior, reports and PDF export, core GUI flows, security controls, and the detection and AI quality metrics.

Out of scope: automated remediation, Windows hosts, Active Directory, real-time monitoring, load testing beyond lab size, and the general quality of third-party AI models beyond the Section 7.2 evaluation in the requirements.

2. Test Approach

Level	What it covers	Tool	When
Unit	One rule, parser, scorer or sanitizer at a time	pytest with fixtures	Every commit
Integration	Collect or import, parse, audit, score and store using snapshots	pytest and the CLI	Before each merge to the main branch
System (lab)	Live collection from the MIT Kerberos lab with planted misconfigurations (TV-01 to TV-15)	CLI and manual checks	End of each sprint
AI evaluation	Explanation and recommendation accuracy, consistency, usefulness	Evaluation script and reviewers	After the AI module exists
UI	Core screens and flows	Manual checklist	Before each demo

Ground truth. Each scenario has an `expected_findings` manifest written and committed before the audit runs. A script compares results to the manifests and computes detection rate, false-positive rate, false-negative rate and classification accuracy.

3. Test Environment

Component	Detail	Status
KDC server	MIT Kerberos on Ubuntu Server LTS in VirtualBox	Established (Sprint 1 reflection)
Kerberos client VM	Generates authentication activity	To confirm
Admin workstation	Ubuntu desktop with Python 3.10+, PyQt6, pytest	Not yet set up
Test principals and fixtures	Planted misconfigurations for TV-01 to TV-15	Not yet created
Mock AI provider	Returns fixed text for offline tests	Not yet built
OpenAI test key	Used only for the AI evaluation	To obtain

4. Entry Criteria, Exit Criteria and Status Values

- **Entry:** the code under test is merged to the main branch, and the fixture for the scenario exists.
- **Exit for Sprint 2:** every Must case for collection, rules and scoring is Pass, and any Fail has a logged defect with an owner.
- **Exit for the project:** all cases are Pass or have an accepted, documented deviation, and the Section 7 metric targets are met.
- **Status values:** Pass, Fail, Blocked (waiting on something else), Not Yet Tested.

5. Test Cases

5.1 Environment and fixtures

ID	Requirement	Steps	Expected result	Status
TC-ENV-01	A-01	Start the KDC VM, create a test user, run `kinit` and `klist` from the client	Ticket granted; `klist` shows a ticket-granting ticket	Not Yet Tested
TC-ENV-02	TV-01 to TV-15	Create the planted principals, policies and config changes for each scenario; capture `kadmin` and log output	Fixtures saved with an `expected_findings` manifest per scenario	Not Yet Tested
TC-ENV-03	FR-COL-08	Export a snapshot from the clean baseline realm (TV-12)	Snapshot folder with JSON, raw files and hashes	Not Yet Tested

5.2 Collection and parsing

ID	Requirement	Steps	Expected result	Status
TC-COL-01	FR-COL-01	Parse principal fixtures for 10 principals	10 Principal objects with all fields populated and kinds classified	Not Yet Tested
TC-COL-02	FR-COL-02	Parse `getpols` and `getpol` fixtures	Policy values equal the fixture values	Not Yet Tested
TC-COL-03	FR-COL-03	Parse `krb5.conf` and `kdc.conf` fixtures containing weak enctypes	Settings extracted with file, section, key and value	Not Yet Tested
TC-COL-04	FR-COL-04	Parse `kadm5.acl` and the justification file	Admin entries with the correct justified flag	Not Yet Tested
TC-COL-05	FR-COL-05	Parse a KDC log fixture with successful and failed requests	Events with timestamp, principal, client, type and result	Not Yet Tested
TC-COL-06	FR-COL-06	Inspect the command allow-list and run collection with a command logger	Only list and get commands run; no modifying command is possible	Not Yet Tested
TC-COL-07	FR-COL-07, NFR-SEC-04, NFR-REL-01	Feed an unreadable file, a truncated log and a malformed principal line	Error recorded per source; remaining stages run; no crash	Not Yet Tested
TC-COL-08	FR-COL-08, FR-COL-09	Export a snapshot, then import it	Import succeeds, hashes verify, findings equal the live run	Not Yet Tested
TC-COL-09	TV-14	Change one raw artifact in a snapshot, then import	Hash mismatch reported and the audit flagged	Not Yet Tested

5.3 Audit rules

ID	Requirement	Steps	Expected result	Status
TC-KC-01	KC-01, TV-01	Users with passwords 91 days and 90 days old	Finding for 91 days only; observed 91,	Not Yet Tested

ID	Requirement	Steps	Expected result	Status
			threshold 90	
TC-KC-02	KC-02, TV-02, FR-AUD-05	Service principals at 181 and 180 days; a user at 181 days	Finding only for the 181-day service principal	Not Yet Tested
TC-KC-03	KC-03, TV-03	Policy minimum length 11 and 12; one principal with no policy	Findings for length 11 and for the no-policy principal; none for 12	Not Yet Tested
TC-KC-04	KC-04, TV-04	Users inactive 181 days and 179 days; one never authenticated, created 200 days ago	Findings for 181 days and for the never-authenticated user	Not Yet Tested
TC-KC-05	KC-05, TV-05	Service principal with no ticket request for 181 days; one with a recent request	Finding for the unused one only	Not Yet Tested
TC-KC-06	KC-06, TV-09	One admin principal missing from the justification file; one listed	Finding for the unlisted admin only	Not Yet Tested
TC-KC-07	KC-07, TV-06	Enable RC4 in `kdc.conf`; set `allow_weak_crypto = true`; add a principal with a 3DES key; also test an AES-only config	Findings for each weak setting and key; none for AES-only	Not Yet Tested
TC-KC-08	KC-08, TV-07	Ticket lifetime of 25 hours and of 24 hours	Finding for 25 hours only	Not Yet Tested
TC-KC-09	KC-09, TV-10	Renewable lifetime of 8 days and of 7 days	Finding for 8 days only	Not Yet Tested
TC-KC-10	KC-10, TV-08	10 failures in 15 minutes; 9 failures; 10 failures spread over 30 minutes	Finding only for 10 failures within 15 minutes	Not Yet Tested
TC-KC-11	KC-11, TV-11	Request burst above the threshold and one below	Finding only for the burst	Not Yet Tested
TC-KC-12	TV-12, FR-AUD-02, NFR-AUD-01	Run the clean baseline snapshot twice	Zero findings both times; identical output	Not Yet Tested

ID	Requirement	Steps	Expected result	Status
TC-KC-13	FR-AUD-07	Run an audit with no AI configured and the network disabled	Audit and scoring complete	Not Yet Tested
TC-KC-14	TV-15	Run the compound scenario (one service principal with RC4, an old password and an unjustified admin entry)	KC-02, KC-06 and KC-07 all reported for that principal	Not Yet Tested

5.4 Risk scoring and storage

ID	Requirement	Steps	Expected result	Status
TC-RSK-01	FR-RSK-01, FR-RSK-02	Score each rule using its baseline factors	Scores match Requirements Section 6.2 (for example KC-07 is 88.5 Critical, KC-02 is 70.0 High)	Not Yet Tested
TC-RSK-02	FR-RSK-02	Score values 24.9, 25, 49.9, 50, 74.9 and 75	Low, Medium, Medium, High, High, Critical	Not Yet Tested
TC-RSK-03	FR-RSK-03	Change one baseline in config and rerun	New score applied; factor values stored with the finding	Not Yet Tested
TC-DB-01	FR-DB-02	Try UPDATE and DELETE on a stored finding; update a review note	Both rejected for the finding; the review note update succeeds	Not Yet Tested
TC-DB-02	FR-AUD-04, FR-DB-01	Run an audit and read a finding back	Record has rule, object, observed value, threshold, timestamp, artifact reference and hash	Not Yet Tested

5.5 AI Gateway

ID	Requirement	Steps	Expected result	Status
TC-AI-01	FR-AI-01, FR-AI-04, FR-AI-06	Run analysis with the mock provider	Fixed text stored and displayed, labelled AI-generated and advisory	Not Yet Tested
TC-AI-02	FR-AI-02, FR-	Capture the outgoing	Only allow-listed fields;	Not Yet Tested

ID	Requirement	Steps	Expected result	Status
	AI-03	payload for a finding on a principal	principal replaced by a token; no raw log, key or description text	
TC-AI-03	TV-13, TM-02	Put prompt-injection text in a principal description and a log message, then run analysis	Text removed before sending; output unaffected; audit completes	Not Yet Tested
TC-AI-04	FR-AI-09	Provider returns invalid JSON, then valid; separately invalid twice	One retry then success stored; two failures give status unavailable	Not Yet Tested
TC-AI-05	FR-RSK-04	Provider reply includes a field that tries to set a score or severity	Field rejected; stored score unchanged	Not Yet Tested
TC-AI-06	FR-AI-05	Disable the network or key, run analysis and export a report	User notified; audit, scores and report complete without AI sections	Not Yet Tested
TC-AI-07	FR-AI-09	Run one finding and inspect the stored output	Model name, prompt version and timestamp stored	Not Yet Tested
TC-AI-08	FR-AI-08, TV-15	Analyze the compound scenario	One related group naming the three findings with a rationale; remediation order shown beside score order	Not Yet Tested

5.6 Reports

ID	Requirement	Steps	Expected result	Status
TC-RPT-01	FR-RPT-01, FR-AI-07	Generate the technical report for the TV-15 audit	Findings, scores, evidence references and AI text present	Not Yet Tested
TC-RPT-02	FR-RPT-02, FR-AI-07	Generate the executive report	Posture summary, counts by severity, top risks and compliance summary	Not Yet Tested

ID	Requirement	Steps	Expected result	Status
TC-RPT-03	FR-RPT-03	Export a PDF to a chosen folder; cancel; export to an unwritable folder	File opens; cancel leaves nothing; clear error for the unwritable folder	Not Yet Tested
TC-RPT-04	FR-RPT-04, FR-RPT-05	Generate a report, edit a finding row in a copy of the database, run verify	Digest mismatch reported for the copy; the original verifies	Not Yet Tested

5.7 User interface

ID	Requirement	Steps	Expected result	Status
TC-UI-01	FR-UI-01, NFR-PERF-02	Start an audit from the GUI in snapshot mode	Progress shown; window stays responsive; results appear	Not Yet Tested
TC-UI-02	FR-UI-02	Open the dashboard with mixed findings; filter by severity and category	Sorted by score; filters work; counts match the database	Not Yet Tested
TC-UI-03	FR-UI-03, FR-UI-04	Open a finding, mark it reviewed and add a note	Detail shows all fields; status and note saved	Not Yet Tested
TC-UI-04	UC-01 alternate flow	Start a live audit with the KDC unreachable	Error shown; audit ends cleanly	Not Yet Tested
TC-UI-05	FR-UI-05	Run the CLI audit on a snapshot	JSON results written; exit code 0	Not Yet Tested

5.8 Security and non-functional

ID	Requirement	Steps	Expected result	Status
TC-SEC-01	NFR-SEC-02	Create the database, a snapshot and a report; check permissions	Owner-only access (600 for files, 700 for folders)	Not Yet Tested
TC-SEC-02	NFR-SEC-03	Scan the repository and config for API keys; run with the key unset	No secrets found; app reports the	Not Yet Tested

ID	Requirement	Steps	Expected result	Status
			missing key and continues without AI	
TC-SEC-03	NFR-SEC-01	Compare the Kerberos database before and after an audit	No change	Not Yet Tested
TC-PERF-01	NFR-PERF-01	Audit a snapshot with 500 principals and 100,000 log lines	Finishes in under 5 minutes excluding AI	Not Yet Tested
TC-USE-01	NFR-USE-01	A new user exports a PDF from a snapshot	Done in 5 actions or fewer	Not Yet Tested
TC-PORT-01	NFR-PORT-01	Install from `requirements.txt` on a clean Ubuntu VM and launch	Installs without manual fixes; application starts	Not Yet Tested
TC-MNT-01	NFR-MNT-01	Run the suite and check each rule has a passing and a failing case	Coverage check passes	Not Yet Tested

5.9 Metric evaluation

ID	Requirement	Steps	Expected result	Status
TC-EVAL-01	Requirements Section 8.1	Run all 15 scenarios and compute metrics against the manifests	Detection rate 95% or higher, false negatives under 5%, false positives under 10%, classification accuracy 95% or higher	Not Yet Tested
TC-EVAL-02	Requirements Section 7.2	Score 55 AI outputs (11 rules x 5 runs) plus TV-15	Accuracy 90% or higher, consistency 95% or higher, usefulness 4.0 or higher	Not Yet Tested

6. Defects and Known Issues

No defects are recorded because no tests have been executed. Defects will be logged here with the test case, a description, a severity and an owner.

Defect	Test case	Description	Severity	Status
None yet				

Known issues that may affect testing:

1. The exact `kadmin.local` field labels and KDC log formats are unverified until fixtures are captured from the lab.
2. Last-authentication data may not support KC-04 as designed; the KDC log may also be needed.
3. The KC-11 default threshold and the OpenAI model are not yet chosen.
4. The Kerberos client VM and the admin workstation are not yet confirmed.

7. Status Summary

Category	Cases	Pass	Fail	Blocked	Not Yet Tested
Environment and fixtures	3	0	0	0	3
Collection and parsing	9	0	0	0	9
Audit rules	14	0	0	0	14
Risk scoring and storage	5	0	0	0	5
AI Gateway	8	0	0	0	8
Reports	4	0	0	0	4
User interface	5	0	0	0	5
Security and non-functional	7	0	0	0	7
Metric evaluation	2	0	0	0	2
Total	**57**	**0**	**0**	**0**	**57**

TC-ENV-01 can be run immediately on the existing KDC VM; recording its result is the first real entry in this plan.